



7 The curve C has equation $y = \frac{x^2 + x - 4}{x^2 + x + 2}$.

(a) State the equation of the asymptote of C . [1]

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(b) Show that, for all real values of x , $-\frac{17}{7} \leq y < 1$. [4]

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(c) Find the coordinates of any stationary points of C . [3]

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(d) Sketch C , stating the coordinates of the intersections with the axes.

[3]

(e) Sketch the graph with equation $y = \frac{|x|^2 + |x| - 4}{|x|^2 + |x| + 2}$ and find the set of values of x for which

$$\frac{|x|^2 + |x| - 4}{|x|^2 + |x| + 2} < -\frac{1}{2}.$$

[5]

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